M-Compatible Gauge Classification for Banach Presheaf Coboundaries

The ℓ^1 Maximal-Symmetry Endpoint

Jeremy H. Carroll

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Abstract

We classify edge-additive obstruction diagnostics on Banach-presheaf 1-cochains under declared local transformation monoids. Specifically, let P be a finite partially ordered set and F a Banach presheaf over P. The space $C^1(P, F)$ is the finite ℓ^1 direct sum of the edge stalks, which is the coproduct in the category of Banach spaces and contractive linear maps. We prove that any edge-additive, faithful diagnostic is an edge-weighted ℓ^1 sum of the native stalk norms when functoriality holds with respect to all bounded linear endomorphisms. A single global scalar follows only after adding edge-uniformity assumptions such as edge-transitive/isomorphic edge data or an explicit normalization across edge types.

By defining the **M-Compatible Algebraic Diagnostics** taxonomy, we demonstrate that the local gauge is rigid only when the active transformation monoid M is the complete set of bounded maps (M_{All}). The result is conditional: maximal bounded-linear local symmetry yields native stalk norms and, with edge-additive aggregation, weighted ℓ^1 over edges; weaker monoids yield larger convex-body families. In particular, CPTP monotonicity alone does not uniquely force the trace norm; the trace norm is selected only after adding the usual operational distinguishability axioms.

In short: *Any additive, faithful diagnostic on Banach presheaf coboundaries that is functorial under arbitrary bounded linear maps (i.e., maximal local symmetry) must be a weighted ℓ^1 sum of native stalk norms. Uniform ℓ^1 requires an additional edge-uniformity or normalization assumption. This result characterizes the maximal-symmetry observer class when aggregation satisfies edge-decomposability and exact disjoint additivity. Relaxing these conditions admits alternative obstruction geometries.*

Conventions. Norms are denoted $\|\cdot\|$ with appropriate subscripts; scalar absolute values are denoted $|\cdot|$.

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1 Setup

1.1 Banach Presheaves over Finite Posets

Let $(P, \leq)$ be a finite partially ordered set. The **covering relations** are the edges of the Hasse diagram:

$$E_{\mathrm{cov}}(P) = \{e = (a, b) : a < b,\ \nexists\, c \text{ with } a < c < b\}. \tag{1}$$

A **Banach presheaf** F over P assigns:

- To each $x \in P$, a Banach space $(F(x), \|\cdot\|_{F(x)})$.

- To each $e = (a, b) \in E_{\text{cov}}$, a bounded linear **restriction map** $r_{b,a} : F(b) \to F(a)$ with $\|r_{b,a}\|_{\text{op}} \leq 1$.

A **section** of F is a family $s = (s(x))_{x \in P}$ with $s(x) \in F(x)$.

1.2 The 1-Cochain Space

For each edge $e = (a, b)$, the **coboundary defect** is:

$$\delta_e(s) = r_{b,a}(s(b)) - s(a) \in F(a). \tag{2}$$

The **1-cochain space** is the finite ℓ^1 Banach direct sum:

$$C^1(P, F) = \bigoplus_{e \in E_{\text{cov}}}^{\ell^1} F(a_e) \tag{3}$$

with norm $\|(\delta_e)_e\|_1 = \sum_e \|\delta_e\|_{F(a_e)}$. This is isometrically isomorphic to the space of coboundary vectors. In the category of Banach spaces and contractive linear maps, this finite ℓ^1 direct sum is the coproduct generated by the edge stalks.

The ℓ^1 coboundary norm of a section s is:

$$I(s) = \sum_{e \in E_{\text{cov}}} \|\delta_e(s)\|_{F(a_e)} = \|(\delta_e(s))_e\|_1. \tag{4}$$

1.3 Seminorms on 1-Cochains

Definition 1.1. A **seminorm on 1-cochains** is a function $\nu : C^1(P, F) \to \mathbb{R}_{\geq 0}$ satisfying $\nu(cx) = |c|\nu(x)$ and $\nu(x + y) \leq \nu(x) + \nu(y)$ for all $x, y \in C^1(P, F)$ and scalars c.

We consider four properties:

(P1) Edge-wise additivity. ν decomposes as

$$\nu((\delta_e)_e) = \sum_{e \in E_{\text{cov}}} \nu_e(\delta_e) \tag{5}$$

where each $\nu_e : F(a_e) \to \mathbb{R}_{\geq 0}$ is itself a seminorm on $F(a_e)$.

(P2) Faithfulness. For each e: $\nu_e(\delta) = 0 \iff \delta = 0$.

(P3) Functoriality under M-Compatible Subsets (Contractive Monotonicity). The function ν_e is compatible with the underlying Banach geometry restricted by an admissible monoid evaluated on the stalk, $M_e \subset \text{End}_b(F(a_e))$. Specifically, for a bounded linear transformation evaluated in this monoid:

$$\nu_e(\varphi(\delta)) \leq \|\varphi\|_{F(a_e) \to F(a_e)} \cdot \nu_e(\delta) \quad \forall \varphi \in M_e. \tag{6}$$

(Note: When $M_e = \{$All bounded linear maps$\}$, this imposes absolute direction independence, forcing proportionality to the native stalk norm. When $M_e = \{$CPTP maps$\}$, monotonicity restricts admissible gauges to quantum distinguishability monotones; the Schatten-1 trace norm requires additional operational assumptions. This classification distinguishes the maximal-linear endpoint from domain-specific relaxations.)

(P4) Isomorphism invariance. For any poset isomorphism $\pi : \tilde{P} \to P$:

$$\nu_{\tilde{P}}(\pi^*(\delta_e)_e) = \nu_P((\delta_e)_e). \tag{7}$$

1.4 Relation to the Combinatorial Classification

The present classification may be viewed as a functional-analytic extension of the finite-dimensional result established in "Coordinate-Wise Additivity and the ℓ^1 Norm on Finite Graph Cochains" [4]. In that setting, ℓ^1 arises from local additivity over disjoint coordinate supports and edge separability alone, with no Banach space structure required. Here, we replace coordinate additivity with structural compatibility in the form of functoriality under bounded linear maps on Banach spaces (P3), showing that the same rigidity persists in a substantially more general setting. The combinatorial result [4] thus provides the minimal case; the present paper demonstrates that the ℓ^1 conclusion is not an artifact of finite coordinates, but survives the passage to infinite-dimensional Banach stalks.

Although numbered Paper 002 in the preprint series, this paper is logically prior to projection-obstruction applications that use its diagnostic classification. The numbering reflects publication order, not logical dependency.

Scope note. The convex-body taxonomy in Sections 2–3 is stated first for finite-dimensional stalks. The maximal-symmetry endpoint in Sections 4–5 applies to Banach stalks using Hahn–Banach. Infinite-dimensional convex-body classifications require the usual topological qualifications and are not claimed in full generality here.

2 General M-Compatible Gauge Classification

2.1 Definition

Let V be a finite-dimensional normed vector space and let

$$M \subset \mathrm{End}_b(V) \tag{8}$$

be an admissible transformation monoid. The operator norm $\|T\|$ is taken with respect to a fixed ambient Banach norm $\|\cdot\|_V$ on V. Thus the M-compatible classification depends on the pair $(M, \|\cdot\|_V)$, not on M alone.

A gauge $j : V \to \mathbb{R}_{\geq 0}$ is **M-compatible** if:

1. $j(x) = 0 \iff x = 0$
2. $j(\lambda x) = |\lambda| j(x)$
3. $j(x + y) \leq j(x) + j(y)$
4. $j(Tx) \leq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} j(x) \quad \forall T \in M$

2.2 Theorem — M-Compatible Gauge / Convex Body Correspondence

Theorem 2.1. *Let $B_j := \{x \in V : j(x) \leq 1\}$. Then j is an M-compatible faithful gauge if and only if B_j is a subset of V satisfying:*

$$B_j \text{ is closed, convex, balanced, absorbing, and containing no nontrivial line} \tag{9}$$

(boundedness over the ambient vector space additionally ensures equivalence with a proper norm), and

$$T(B_j) \subseteq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} B_j \quad \forall T \in M. \tag{10}$$

Conversely, every closed, convex, balanced, absorbing body containing no nontrivial line $B \subset V$ satisfying

$$T(B) \subseteq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} B \tag{11}$$

defines an M-compatible gauge by its Minkowski functional:

$$j_B(x) := \inf\{\lambda > 0 : x \in \lambda B\}. \tag{12}$$

Thus: $\mathrm{Gauge}(V, M) \cong \{\text{M-contractive symmetric convex bodies in } V\}$.

2.3 Proof Sketch

Proof Sketch. The gauge axioms imply that B_j is closed, convex, balanced, and absorbing. Faithfulness excludes nontrivial lines. The compatibility condition gives: $x \in B_j \implies j(x) \leq 1$ so $j(Tx) \leq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} j(x) \leq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V}$. Hence: $Tx \in \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} B_j$.

Conversely, given such a body B, its Minkowski functional is homogeneous and subadditive. The inclusion $T(B) \subseteq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} B$ implies $j_B(Tx) \leq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} j_B(x)$. So j_B is M-compatible. □

2.4 Corollaries

Corollary 1 — Order Reversal. If $M_1 \subseteq M_2$, then $\mathrm{Gauge}(V, M_2) \subseteq \mathrm{Gauge}(V, M_1)$. More admissible transformations mean fewer admissible gauges.

Corollary 2 — Maximal Symmetry Endpoint. If $M = \mathrm{End}_b(V)$ and compatibility means $j(Tx) \leq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} j(x)$ for all $T \in \mathrm{End}_b(V)$, then every compatible gauge is proportional to the native norm: $j(x) = c\|x\|_V$, where $\|\cdot\|_V$ is the ambient Banach norm defining the operator norm. This is the specific local endpoint used in Section 4. Global ℓ^1 over edges enters only after imposing additive edge aggregation.

Corollary 3 — Compact Group Construction. If $G \subset \mathrm{GL}(V)$ is compact and acts by isometries, then every G-invariant symmetric convex body defines a compatible gauge. Given any seed norm q, one can construct a G-invariant norm by averaging: $j_G(x) := \int_G q(gx)\, dg$. So compact symmetry usually produces a family, not a unique gauge.

Corollary 4 — Transitive Sphere Action. If G acts transitively on the native unit sphere of V, then every G-invariant compatible norm is proportional to the native norm: $j(x) = c\|x\|_V$. For example, provided j is a norm (convex, homogeneous), $\mathrm{SO}(k)$-invariance forces Euclidean structure ($j(x) = c\|x\|_2$) when acting on $\mathbb{R}^k$. Passing this local vector geometry through replica-extensive global aggregation natively yields $J = c\ell^1(E; \ell^2)$.

3 Specific M-Invariant Convex Body Classifications

3.1 $M = \mathrm{SO}(k)$ on $\mathbb{R}^k$

Theorem (Rotational symmetry forces Euclidean balls). Let $V = \mathbb{R}^k$ with the Euclidean ambient norm and let $M = \mathrm{SO}(k)$. If j is a norm satisfying $j(Rx) = j(x)$ for all $R \in \mathrm{SO}(k)$, then $j(x) = c\|x\|_2$. Equivalently, the only $\mathrm{SO}(k)$-invariant symmetric convex bodies are Euclidean balls $B = cB_2^k$.
Reason: $\mathrm{SO}(k)$ acts transitively on the Euclidean unit sphere. Therefore j is constant on every Euclidean sphere. Homogeneity gives proportionality to $\|x\|_2$.
Global obstruction: Replica-extensive aggregation gives $J(\delta) = \sum_e c_e \|\delta_e\|_2$. With edge-transitivity: $J = c\ell^1(E; \ell^2)$.

3.2 Signed permutation group on $\mathbb{R}^k$

Let $M = \{\text{signed coordinate permutations}\}$.
Theorem (Signed permutations classify symmetric unconditional bodies). The M-invariant convex bodies are exactly symmetric unconditional convex bodies invariant under coordinate permutation. Equivalently, compatible gauges are symmetric unconditional norms $j(x) = j(|x|)$ and $j(x_1, \ldots, x_k) = j(x_{\pi(1)}, \ldots, x_{\pi(k)})$.
Examples: $\|x\|_1$, $\|x\|_2$, $\|x\|_\infty$, plus symmetric polyhedral norms and Orlicz-type gauges.
Meaning: Signed-permutation symmetry does **not** force ℓ^1 or ℓ^2. It permits an entire family.

3.3 Diagonal contractions on $\mathbb{R}^k$

Let $M = \{\mathrm{diag}(a_1, \ldots, a_k) : |a_i| \leq 1\}$.
Theorem (Diagonal contractions classify monotone unconditional bodies). A convex body B is M-contractive iff its gauge j_B is coordinatewise monotone unconditional: $|x_i| \leq |y_i| \; \forall i \implies j_B(x) \leq j_B(y)$. Equivalently: $x \in B, |y_i| \leq |x_i| \; \forall i \implies y \in B$.
Examples: $\sum_i w_i|x_i|$, $(\sum_i w_i|x_i|^p)^{1/p}$, $\max_i w_i|x_i|$, Orlicz norms, Lorentz norms, weighted boxes, and unconditional polytopes.
Meaning: Diagonal contraction symmetry allows many non-ℓ^1 geometries. It forces monotonicity, not a single norm.

3.4 Positive cone maps

Let $V = \mathbb{R}^k$ with cone $\mathbb{R}^k_{\geq 0}$, and let M be positive/order-preserving maps.
Theorem (Positive maps classify order-compatible gauges). Compatible convex bodies are those whose gauge respects the order $0 \leq x \leq y \implies j(x) \leq j(y)$ on the positive cone, with symmetric extension if signed defects are allowed.
Examples: Weighted ℓ^1, weighted ℓ^∞, monotone lattice norms, base norms on ordered vector spaces.
Meaning: Order-compatible obstruction theory measures defect relative to cone structure, not rotational or full linear symmetry.

3.5 Unitary conjugation on matrices

Let $V = M_n(\mathbb{C})$ or Hermitian matrices, and $M = \{A \mapsto UAU^\dagger : U \in \mathrm{U}(n)\}$.
Theorem (Unitary conjugation classifies spectral convex bodies). The M-invariant convex bodies are exactly spectral convex bodies: membership depends only on eigenvalues for Hermitian matrices, or singular values for general matrices. Compatible gauges have the form $j(A) = g(s(A))$, where $s(A)$ is the singular-value vector and g is a symmetric gauge function.
Examples: Schatten norms ($\|A\|_{S_p} = (\sum_i s_i(A)^p)^{1/p}$), operator norm ($\|A\|_{\mathrm{op}} = s_1(A)$), trace norm ($\|A\|_1 = \sum_i s_i(A)$), Ky Fan norms.
Meaning: Unitary symmetry alone does **not** force trace norm. It forces spectral dependence.

3.6 CPTP maps on Hermitian trace-class differences

Let V be Hermitian trace-zero operators and let M be CPTP maps.
Theorem (CPTP compatibility gives distinguishability monotones). A compatible gauge must satisfy $j(\Lambda X) \leq j(X)$ for all CPTP maps Λ. Thus admissible convex bodies are those invariant under quantum-channel contraction.
Canonical example: Trace norm $j(X) = \frac{1}{2}\|X\|_1$. This is operationally distinguished by Helstrom binary discrimination.
Warning: CPTP monotonicity alone does **not** uniquely force trace norm unless additional axioms are imposed (classical restriction = total variation, binary discrimination duality, tensor stability, effect-operator duality).
Meaning: CPTP selects quantum distinguishability geometry, but uniqueness requires operational assumptions beyond monotonicity.

3.7 No nontrivial monoid

Let $M = \{I\}$.
Theorem (No symmetry admits all gauges). Every norm/gauge is compatible. Convex bodies are all closed, convex, balanced, absorbing bodies containing no nontrivial lines natively.
Meaning: Without symmetry, no canonical obstruction geometry exists.

4 The Maximal Symmetry Endpoint

The theorem in this section classifies diagnostics inside the edge-additive observer class only. Without (P1), no ℓ^1 conclusion follows; mixed, max-type, and cross-edge-coupled diagnostics remain admissible. We first isolate a primary structural lemma.

Lemma 4.1 (Direction Independence under Maximal Symmetry)**.** *If ν_e satisfies (P3) with $M_e = \mathrm{End}_b(F(a_e))$, then for any $v, w \in F(a_e)$ with $\|v\| = \|w\| \neq 0$, we have $\nu_e(v) = \nu_e(w)$.*

Proof. By the Hahn–Banach theorem, there exists $f_v \in F(a_e)^*$ with $\|f_v\|_{\mathrm{op}} = 1$ and $f_v(v) = \|v\|$. Define $\Phi : F(a_e) \to F(a_e)$ by $\Phi(x) = f_v(x)\frac{w}{\|w\|}$. Because Φ is a bounded linear endomorphism of $F(a_e)$, (P3) applies directly here because $M_e = \mathrm{End}_b(F(a_e))$. Then $\Phi(v) = w$ and $\|\Phi\|_{\mathrm{op}} = 1$, so $\nu_e(w) \leq \nu_e(v)$ by (P3). By symmetry, constructing $\Psi(x) = f_w(x)\frac{v}{\|v\|}$ gives $\nu_e(v) \leq \nu_e(w)$, forcing equality. □

Theorem 4.2 (Free ℓ^1 Seminorm Classification)**.** *Let ν be a seminorm on $C^1(P, F)$ satisfying (P1), (P2), and (P3) with $M_e = \mathrm{End}_b(F(a_e))$ for all edges e. Then for each edge e, there exists $w_e > 0$ such that*

$$\nu_e(\delta) = w_e \cdot \|\delta\|_{F(a_e)} \quad \textit{for all } \delta \in F(a_e). \tag{13}$$

If additionally (P4) holds and all edge presheaf data lie in a single isomorphism orbit (or an explicit cross-edge normalization is imposed), then $w_e = \alpha$ is independent of e, and

$$\nu((\delta_e)_e) = \alpha \sum_e \|\delta_e\|_{F(a_e)} = \alpha \cdot I(s). \tag{14}$$

In other words: maximal local functoriality gives a weighted ℓ^1 coboundary norm. The unweighted ℓ^1 coboundary norm is unique up to positive scalar only under the additional edge-uniformity hypothesis.

5 Proof of the Maximal Theorem

The proof proceeds in three stages.

Proof. **Stage 1: Direction Independence.** This is Lemma 4.1: for any two vectors $v, w \in F(a_e)$ with $\|v\| = \|w\| \neq 0$, we have $\nu_e(v) = \nu_e(w)$.

Stage 2: Each ν_e is proportional to $\|\cdot\|$. Fix an edge e and let $V = F(a_e)$. By Lemma 4.1, all unit vectors in V have the same ν_e-value: there exists $w_e \geq 0$ such that $\nu_e(v) = w_e$ for all v with $\|v\| = 1$. By (P2), $w_e > 0$. For any $\delta \in V$ with $\delta \neq 0$, the absolute homogeneity of ν_e gives:

$$\nu_e(\delta) = \|\delta\| \cdot \nu_e\left(\frac{\delta}{\|\delta\|}\right) = \|\delta\| \cdot w_e = w_e\|\delta\|. \tag{15}$$

Since $\nu_e(0) = 0$ by the seminorm property, we conclude $\nu_e(\delta) = w_e\|\delta\|$ for all $\delta \in V$.

Stage 3: Isomorphism invariance forces uniform weights. Because of (P1), the seminorm **splits as a direct sum**, so each component can be evaluated in isolation. Let $e = (a, b)$ be any covering edge of a finite poset P. Define the single-edge subposet $P_e = \{a < b\}$. Each summand $\nu_e(\delta_e)$ depends only on the coboundary defect $\delta_e(s) = r_{b,a}(s(b)) - s(a)$, which is computed entirely from the restricted presheaf data $(F(a), F(b), r_{b,a})$. The value $\nu_e(\delta_e)$ is therefore intrinsically independent of the ambient poset P in which e is embedded: it is determined by the single-edge presheaf (P_e, F_e).

By Stages 1–2, $\nu_e(\delta_e) = w_e\|\delta_e\|$ for some $w_e > 0$. Since w_e depends only on the isomorphism class of the single-edge presheaf (P_e, F_e), it remains to show all such presheaves yield the same weight. Let $e_1 = (a_1, b_1)$ and $e_2 = (a_2, b_2)$ be any two covering edges with the same stalk spaces $F(a_{e_1}) \cong F(a_{e_2})$ and $F(b_{e_1}) \cong F(b_{e_2})$, and compatible restriction maps under the induced

isomorphism. The single-edge posets $P_{e_1} = \{a_1 < b_1\}$ and $P_{e_2} = \{a_2 < b_2\}$ admit a unique order-preserving bijection $\pi : P_{e_1} \to P_{e_2}$. By (P4), $\nu_{P_{e_1}}(\pi^*(\delta_{e_2})) = \nu_{P_{e_2}}((\delta_{e_2}))$, which gives $w_{e_1} = w_{e_2}$.

Thus all covering edges in the same edge-presheaf isomorphism orbit share the same weight. If the paper additionally assumes a single edge orbit or a global normalization identifying all edge types, set $\alpha = w$ and obtain $\nu = \alpha \cdot I$. Without that extra assumption the correct conclusion is the orbit-weighted form

$$\nu((\delta_e)_e) = \sum_{\mathcal{O}} w_{\mathcal{O}} \sum_{e \in \mathcal{O}} \|\delta_e\|. \tag{16}$$

□

6 Necessity of Each Property

Each property is independently necessary. Together they are sufficient.

Table 1: Necessity of Diagnostic Axioms

Dropped property	What happens	Example
(P1) Additivity	ℓ^∞-type norms admitted	$\nu = \max_e \|\delta_e\|$
(P2) Faithfulness	Trivial seminorm admitted	$\nu \equiv 0$
(P3) Functoriality	Direction-dependent seminorms admitted	On $F(a_e) = \mathbb{R}^2$: $\nu_e(x_1, x_2) = \|x_1\| + 2\|x_2\|$; satisfies (P1), (P2), (P4), but violates direction independence (Lemma 4.1), hence is not proportional to $\|\cdot\|$
Edge-uniformity	Orbit-dependent weights admitted	$\nu = \sum_{\mathcal{O}} w_{\mathcal{O}} \sum_{e \in \mathcal{O}} \|\delta_e\|$

7 Structural Uniqueness

Rather than defining artificial categories to encapsulate simple scaling relations, we formulate the universal structure directly:

Proposition 7.1 (Normalized Uniqueness)**.** *In the maximal-local-symmetry setting $M_e = \mathrm{End}_b(F(a_e))$, the weighted ℓ^1 coboundary norm is the unique additive diagnostic satisfying properties (P1)–(P3). Adding edge-uniformity or single-orbit isomorphism invariance reduces the weighted form to $\alpha \cdot I(s)$ for some positive scalar α.*

Corollary 7.2 (Commutative Factorization)**.** *For any seminorm ν satisfying (P1)–(P4) and any admitted bounded linear map φ acting stalkwise with the declared operator-norm bound, the evaluation functional factors predictably:*

$$\nu(\varphi_*(\delta)) \leq \|\varphi\|_{op} \cdot \nu(\delta), \tag{17}$$

and, in the maximal-local-symmetry endpoint, the weighted ℓ^1 coboundary norm is the unique edge-additive framework for which this structural compatibility with all bounded linear maps holds. The unweighted scalar form is recovered only after edge-uniformity or normalization.

8 Discussion

8.1 What This Paper Proves

This paper isolates a purely structural constraint on how coboundary defects in Banach presheaf systems can be measured: once one requires edge-wise additivity, faithfulness, and absolute compatibility with the ambient Banach geometry through functoriality under *all* bounded linear maps, the resulting seminorm is limited to a weighted ℓ^1 sum of native stalk norms. A global scalar is a further edge-uniformity conclusion, not a consequence of local functoriality alone.

Crucially, this uniqueness is tightly bounded by the choice of the transformation monoid M. The classification does not claim that physics universally employs ℓ^1. Rather, restricting to a narrower class of transformations generates the **M-Compatible Diagnostics**. Specific monoids classify specific convex-body geometries. Maximal bounded-linear symmetry collapses the admissible body class to the native Banach ball.

8.2 Relation to Other Work

The Hahn–Banach factoring argument used in Stage 1 is analogous to classical norm classification techniques in Banach space theory. The result is a presheaf version of a classical principle in Banach space theory: the characterization of the ℓ^1 norm as the unique additive cross-norm on a finite Banach sum.

8.3 Strategic Place in the Overall Program

In the logical dependency order, this paper supplies the maximal-symmetry gauge-classification input used by later projection and Hodge-style obstruction papers. It sits near the start of the sequence establishing conditional ℓ^1 rigidity for additive, structure-preserving measurements of local defects.

8.4 Connection to Observer Theory and Contextual Congruence

The M-compatible gauge classification developed here determines the **diagnostic geometry** of the obstruction calculus. The general finite obstruction framework [6] adds a complementary **governance layer** that determines what authority that geometry carries.

Observer Passivity and Gauge Choice

The gauge choice studied here — the selection of local norms via the M-compatible classification — determines *how* the quotient obstruction Φ is computed. The observer passivity theory then constrains *what happens when that diagnostic is further processed.*

Contextual Congruence and Lift Triggers

The general framework shows that the lift trigger is not obstruction mass ($\Phi > 0$) alone but **context-visible non-congruence**. The M-classification tells us *what geometry to use*; contextual congruence tells us *when to act on it.*

Reflective Completion and Sheaf Gluing

The diagnostic computed by the M-selected gauge feeds into the completion procedure: each witness exposes a nonzero quotient residual, and the selected norm determines the quantitative obstruction mass used in minimality certificates. The Banach presheaf coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ studied here is exactly the local-to-global defect that appears in the sheaf gluing obstruction theorem.

8.5 Norm Choice and Obstruction Geometry

Let

$$\Phi(\omega) = \inf_f \|\omega - d_0 f\|_J \tag{18}$$

be the observer-induced obstruction functional. The choice of norm J determines the geometry of residuals. Under additional maximal-observer assumptions (additivity, refinement invariance, regularity, maximal bounded-linear local functoriality, edge-uniformity), the functional Φ is the unweighted ℓ^1 endpoint: $\Phi(\omega) = C \sum |h_{1,i}|$.

8.6 Connection to Dynamical Obstruction Theory

The classification result established here is purely structural. Its significance becomes more apparent when combined with dynamical settings in which such coboundary defects are generated. The ℓ^1 diagnostic appearing in projection obstruction is not an unconstrained modeling choice; it is selected by the stated observer and symmetry assumptions.

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References

1. V. Pestov, "Dynamics of infinite-dimensional groups and Ramsey-type phenomena," *Publicacoes Matematicas do IMPA*, 2006.

2. R. E. Megginson, *An Introduction to Banach Space Theory*, Springer, 1998.

3. J. H. Carroll, "Projection Obstruction for Factorized States: Nonlinear Dynamics, Conditional ℓ^1 Diagnostics, and Finite-Time Defect Bounds," Zenodo, 2026. https://zenodo.org/records/201726

Changelog

v4.0 (May 2026): Retitled the paper around M-compatible gauge classification for Banach presheaf coboundaries. Clarified finite-dimensional convex-body scope versus Banach-stalk maximal-symmetry theorem. Renumbered internal results.
v3.0 (May 2026): Added §8.4: Connection to Observer Theory and Contextual Congruence. Linked M-compatible gauge classification to observer passivity (O1), contextual congruence, reflective completion, and sheaf gluing obstruction.
v2.0 (April 2026): Initial pre-print with M-compatible classification, maximal symmetry endpoint, and necessity analysis.